

PartForge

Collect and trade rocket component NFTs, then assemble them into working ships

The Problem & Our Solution

NFT games rarely let collectibles affect real gameplay. PartForge makes rocket part NFTs the actual engine of play.

- Most NFT games treat items as static collectibles with no real gameplay impact
- Players can't verify that in-game outcomes are actually tied to what they own
- PartForge encodes rocket parts, engines, tanks, fins, as NFTs with on-chain stats
- Flight simulation results are computed directly from owned part metadata
- Trading parts becomes strategically meaningful, not just cosmetic collecting



How It Works

Players mint, trade, and assemble rocket part NFTs. The simulator reads on-chain metadata to compute real flight performance.

- Mint individual parts (engine, tank, fin) as NFTs with on-chain stats via Metaplex
- Inventory UI lets players select owned parts and assemble a custom rocket
- Simulation engine reads combined part metadata on-chain to compute flight results
- Marketplace enables buying, selling, and trading individual part NFTs
- Anchor programs on Solana handle minting, assembly logic, and marketplace trades

Built with

Solana, Anchor, Metaplex Token Metadata, React, Node.js backend, Phantom Wallet Adapter

Why Now & What's Next

Composable NFT standards on Solana finally make metadata-driven gameplay practical. We're building toward deeper crafting and competition.

- Solana's Metaplex Token Metadata standard makes on-chain item data easy to read in gameplay
- This enables verifiable, skill-based outcomes instead of purely cosmetic NFT ownership
- Next: crafting and fusion mechanics to combine parts into rarer ones
- Next: seasonal part drops and limited edition engines to refresh the marketplace
- Next: leaderboard tying part quality and build skill to competitive rankings

Category

Gaming